Autosomal Dominant Nebulin-Related Myopathy (MONDO:1010152): A Comprehensive Disease Characterization

Summary

Autosomal Dominant Nebulin-Related Myopathy (MONDO:1010152) is a rare, mild, slowly progressive, predominantly **distal** skeletal muscle myopathy caused by heterozygous pathogenic variation in **NEB**, the gene on chromosome 2q23.3 encoding **nebulin** — a giant (~700–800 kDa) filamentous sarcomeric thin-filament protein that acts as a "molecular ruler" to specify and stabilize actin thin-filament length. The dominant distal form is specifically caused by **single large heterozygous (sometimes mosaic) multi-exon deletions spanning ~52–97 exons of NEB**. This contrasts sharply with the far more common autosomal **recessive** nemaline myopathy type 2 (NEM2; OMIM 256030), which arises from biallelic (compound heterozygous or homozygous) NEB variants and produces a more severe, proximal-predominant congenital phenotype.

The mechanistic chain is well established at the molecular level: loss/dysfunction of nebulin leads to **shortened and non-uniform actin thin filaments**, **reduced myofilament calcium sensitivity**, **altered crossbridge cycling kinetics**, and a **destabilized myosin super-relaxed (SRX) state** that raises resting ATP consumption. Together these produce a ~60% loss of force-generating capacity and the characteristic muscle weakness. A key unresolved question is whether the dominant large deletions act via **haploinsufficiency** or a **dominant-negative** mechanism — the curated MONDO definition asserts a dominant-negative mechanism, but direct experimental proof for the dominant distal form specifically is still lacking.

There is **no curative therapy**. Management is supportive (respiratory support, physiotherapy/orthotics for foot drop, feeding and bulbar support, orthopedic care). The most mechanistically rational investigational approaches are **fast skeletal troponin activators** (e.g., CK-2066260, which restores calcium sensitivity of force in patient fibers) and **myosin modulators** (e.g., Mavacamten, which reverses ATP overconsumption in vitro). This report synthesizes 10 confirmed findings from 48 reviewed papers into a complete disease-knowledge-base entry across all 15 requested domains.

Key Findings

Finding 1 — Large heterozygous NEB deletions cause a mild dominant distal phenotype

The defining molecular lesion of this disease is a **single large heterozygous deletion of NEB**. In the landmark cohort of Sagath et al. (2025), 35 families with NEB structural variants were studied; among them, **12 families with a distal myopathy phenotype** (10 previously unpublished) carried **8 unique large deletions encompassing 52–97 exons**, present in either heterozygous (n=10) or mosaic (n=2) state. Critically, *"all patients with large dominant deletions in NEB had milder, predominantly distal muscle weakness,"* and the authors state that *"large deletions have been shown to cause dominantly inherited distal myopathies"* PMID: 40517164. This established, for the first time, a statistically significant association between large heterozygous NEB deletions and a dominant distal phenotype, cleanly separating this entity from recessive NEM2.

Finding 2 — NEB encodes nebulin, a giant thin-filament molecular ruler

Nebulin (~700–800 kDa) is a giant filamentous protein that spans nearly the entire length of the actin thin filament in skeletal muscle sarcomeres. Cryo-EM/tomography of intact myofibrils demonstrated that *"in skeletal muscle, nebulin stabilizes and regulates the length of thin filaments"* and that the structures *"support the role of nebulin as a thin filament 'molecular ruler' and provide a molecular basis for studying nemaline myopathies"* PMID: 35175800. Nebulin interacts with actin and a troponin-T linker (not directly with myosin or tropomyosin), binds CapZ at the Z-disc to anchor thin-filament barbed ends PMID: 18272787, and regulates muscle contraction, Z-disc formation, and myofibril assembly PMID: 26792324.

Finding 3 — Nebulin deficiency weakens muscle via shortened filaments, reduced Ca^{2+} sensitivity, and altered crossbridge cycling

In NEB-mutant patient muscle, nebulin is severely reduced (especially its N-terminus), force-generating capacity falls by ~60%, and the force–sarcomere length relationship is left-shifted. Average thin-filament length is *"reduced from approximately 1.3 microm in control muscle to approximately 0.75 microm in NM-NEB muscle"* PMID: 19346529. Beyond filament length, nebulin-deficient fibers *"develop less force due to reduced myofilament calcium sensitivity and altered crossbridge cycling kinetics"* PMID: 21115852. Conditional Neb-knockout mice

recapitulate reduced specific force from fewer force-generating crossbridges and shorter thin filaments, with muscle-type-dependent atrophy via MuRF-1/MUSA1/Gadd45a proteolysis PMID: 26123491. A fast skeletal troponin activator, CK-2066260, *"greatly increased the calcium-sensitivity of force generation"* in patient fibers PMID: 23572184.

Finding 4 — NEB gene architecture and the recessive-vs-dominant divide

NEB (HGNC-approved symbol **NEB**; chromosome **2q23.3**) is one of the largest human genes with ~183 exons, including an 8-exon triplicate region (exons 82–105) and alternatively spliced exons. **Recessive NEM2 (OMIM 256030)** results from biallelic variants — point mutations, small indels, splice variants, and intragenic copy-number changes. The in-frame deletion of **exon 55 (NEB ΔExon55)** is the single most common disease-causing variant and a **founder mutation in Ashkenazi Jewish patients** — a model was designed to *"replicate a founder mutation seen frequently in patients with nemaline myopathy with Ashkenazi Jewish heritage"* PMID: 23715096. By contrast, single large heterozygous deletions spanning 52–97 exons cause the autosomal dominant distal myopathy: *"eight unique, large deletions encompassing 52-97 exons in either heterozygous (n = 10) or mosaic (n = 2) state were identified"* PMID: 40517164.

Finding 5 — Epidemiology, clinical, and diagnostic profile

Nemaline myopathy overall affects **~1 in 50,000 live births**, with prevalence up to ~1 in 20,000 in certain populations; NEB is the most common causative gene (>50% of cases) PMID: 31530540. NEB-associated phenotypes span classic nemaline myopathy, core-rod myopathy, and distal myopathies — *"pathogenic NEB variants have been identified in core-rod myopathy and in distal myopathies"* PMID: 25205138. Diagnosis rests on muscle biopsy (nemaline rods — Z-disc-derived electron-dense bodies), genetic testing, EMG (myogenic pattern), and muscle MRI. In NEB-related NM, *"most patients (18/21) presented with lower limb muscle weakness, with a distal predominance,"* and MRI shows greatest fatty infiltration in soleus, gastrocnemius, and gluteus maximus PMID: 40583855. Natural history is relatively stable; in a cross-sectional cohort, 58% needed mechanical support and 65% had abnormal pulmonary function PMID: 33397769.

Finding 6 — Animal and cellular models recapitulate the disease

Multiple models reproduce NEB myopathy: conditional Neb-KO mice develop nemaline rods, <5% nebulin, shortened thin filaments, reduced specific force, and fiber-type switching PMID: 26123491; Neb ΔExon55 mice replicate the Ashkenazi founder mutation, producing *"nemaline bodies—a hallmark feature of nemaline myopathy—in muscle fibres from Neb(ΔExon55) mice"*

PMID: 23715096; and an allelic series of zebrafish neb mutants *"recapitulate most aspects of NEB-based NM, showing drastically reduced survival, defective muscle structure, reduced contraction force, shorter thin filaments"* PMID: 38493359.

Finding 7 — No curative therapy; supportive management with mechanism-based leads

Management is primarily supportive: respiratory support (NIV/ventilation), physiotherapy/orthotics for distal weakness and foot drop, feeding/bulbar support, and orthopedic care — *"management remains mainly supportive, and currently, there is no available curative treatment"* PMID: 38866581. Controlled model studies of nutritional supplements (L-tyrosine, L-carnitine, taurine, creatine) *"revealed no significant improvement in skeletal muscle function emphasizing the urgency to develop effective therapies for nemaline myopathy"* PMID: 29848386. Mechanism-based leads include fast skeletal troponin activators and myosin modulators.

Finding 8 — Variant architecture determines inheritance and severity

In the 35-family cohort, recessive disease arose from compound-heterozygous intragenic deletions/duplications or triplicate-region gains plus a small trans variant (23 families; 2 homozygous), and *"in the families where inheritance was recessive, no correlation could be made between the types of variants and the severity of the disease."* In contrast, the 12 distal-myopathy families each carried a single large heterozygous deletion, and the authors *"establish a clear and statistically significant association"* between large dominant NEB deletions and the distal phenotype PMID: 40517164. Independent large cohorts (de Feraudy et al., 49 nebulin-related congenital myopathy patients) confirm the histological spectrum of rods ± cores PMID: 42609597.

Finding 9 — NEB mutations destabilize the myosin super-relaxed state and remodel the energy proteome

In muscle fibers from NEB-NM patients and nebulin-deficient mice, *"the myosin stabilizing conformational state, known as super-relaxed state, was significantly impaired, inducing an increase in the energy (ATP) consumption of resting muscle fibres from NEB-NM patients,"* which helps explain why *"histological analyses of patients' muscle biopsies often reveal unexplained accumulation of glycogen and abnormally shaped mitochondria"* PMID: 36528760. The myosin modulator Mavacamten (promotes SRX) reversed myosin ATP overconsumption in vitro in patient fibers, but 4-week in vivo treatment of conditional Neb-KO mice did not correct the dysregulated energy proteome PMID: 39216086.

Finding 10 — MONDO curation confirms gene, phenotype, and asserted mechanism

The MONDO ontology entry **MONDO:1010152** (label "autosomal dominant nebulin-related myopathy"; cross-reference GARD:0028130) defines the disease as *"any myopathy in which an autosomal dominantly inherited genetic variation in the NEB gene causes disease via a dominant-negative mechanism,"* with symptoms including *"distal muscle weakness, hypotonia, muscle fiber atrophy, foot drop, high arched palate, feeding difficulties, and type 1 fiber predominance."* This confirms NEB causation and the dominant, distal-predominant phenotype.

Full Section-by-Section Report

1. Disease Information

Overview. Autosomal dominant nebulin-related myopathy is a rare inherited skeletal muscle disease caused by **heterozygous large multi-exon deletions of the NEB gene**. It presents as a **mild, slowly progressive distal myopathy** — distinct from the severe, proximal, congenital recessive nemaline myopathy type 2 (NEM2) caused by biallelic NEB variants.

Key identifiers. - **Mondo:** MONDO:1010152 (autosomal dominant nebulin-related myopathy) - **Causal gene:** NEB (HGNC:7720), chromosome 2q23.3 - **GARD:** 0028130 (cross-reference in MONDO) - **Related OMIM:** NEM2 / 256030 (recessive nebulin nemaline myopathy); NEB gene OMIM 161650 - **ICD-11:** 8C70 (Primary disorders of muscles) / congenital myopathy category; **ICD-10:** G71.2 (congenital myopathies) - **MeSH:** Myopathies, Nemaline (D017696) - **Orphanet:** Nemaline myopathy (ORPHA:607) and distal myopathy subtypes

Synonyms / alternative names: NEB-related dominant distal myopathy; nebulin-related distal myopathy; distal nemaline/nebulin myopathy. (Formal synonyms "not recorded" per the template.)

Information source: Aggregated disease-level resources (OMIM, Orphanet, MONDO, HPO) combined with primary cohort and case-level literature (not EHR-derived).

2. Etiology

Disease causal factors. Purely **genetic** — heterozygous large multi-exon deletions (~52–97 exons) of NEB. No environmental, infectious, or acquired cause. (Note: an acquired, non-genetic mimic, sporadic late-onset nemaline myopathy [SLONM], is a separate entity frequently associated with monoclonal gammopathy —

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— and must not be confused with this inherited disease.)

Genetic risk factors. The causal variant class is the large heterozygous NEB deletion. Mosaicism (n=2/12 families) can also cause disease PMID: 40517164. No susceptibility loci or validated modifier genes are established for the dominant form; a modifier screen in recessive models failed to identify modifiers PMID: 31530540.

Environmental / lifestyle risk and protective factors: None established. No protective variants or alleles identified.

Gene–environment interactions: None documented.

3. Phenotypes

Phenotype	Type	HPO term	Onset / severity / frequency
Distal muscle weakness (predominant)	Clinical sign	HP:0002460	Childhood–adult; mild; distal predominance (~18/21 NEB-NM lower-limb cases)
Foot drop	Clinical sign	HP:0009027	Chronic, slowly progressive
Muscle fiber atrophy	Pathology	HP:0009130	Variable
Hypotonia	Clinical sign	HP:0001252	Neonatal/childhood in congenital forms
High-arched palate	Physical	HP:0000218	Congenital
Feeding difficulties	Symptom	HP:0011968	Infancy (more severe forms)
Type 1 fiber predominance	Lab/ pathology	HP:0003803	Biopsy finding
Respiratory insufficiency	Clinical sign	HP:0002093	Variable; 65% abnormal PFTs across NM cohort
Bulbar dysfunction	Clinical sign	HP:0001283	Variable
Nemaline rods on biopsy	Lab/ pathology	HP:0003798	Diagnostic hallmark

Characteristics: In the dominant distal form specifically, onset is typically childhood–adult, severity is **mild**, and progression is **slow/relatively stable** PMID: 40517164; PMID: 33397769. Distal lower-limb weakness with foot drop dominates. **Quality-of-life impact:** foot drop and distal weakness impair gait and daily mobility; respiratory and bulbar involvement, when present, drive the greatest QoL burden. Formal EQ-5D/SF-36 data for the dominant subtype are not available.

4. Genetic / Molecular Information

Causal gene: NEB (HGNC:7720; OMIM 161650), 2q23.3, ~183 exons, one of the largest human genes. Contains an 8-exon triplicate region (exons 82–105) and alternatively spliced exons.

Pathogenic variants (dominant form): Single **large heterozygous deletions** spanning **52–97 exons** (structural variants / large copy-number losses), classified pathogenic; 10/12 heterozygous, 2/12 mosaic PMID: 40517164. **Variant class:** multi-exon structural deletion (CNV). **Functional consequence:** loss of a large internal nebulin segment; mechanism debated between **haploinsufficiency** and **dominant-negative** (MONDO asserts dominant-negative).

Recessive contrast (NEM2): biallelic point mutations, small indels, splice variants, intragenic deletions/duplications, and triplicate-region gains; pathogenic variants represent only ~7% of NEB coding variants PMID: 25205138. The **exon 55 in-frame deletion** is the most common single disease variant and an Ashkenazi Jewish founder mutation PMID: 23715096.

Allele frequency: Large pathogenic NEB deletions are extremely rare/absent in gnomAD. **Origin:** germline (with documented mosaicism). **Modifier genes / epigenetics / chromosomal abnormalities:** none established for the dominant form.

5. Environmental Information

Not applicable — this is a monogenic disorder with no established environmental, lifestyle, or infectious contributors. (Acquired nemaline rod pathology exists as separate entities: SLONM/MGUS-associated and inflammatory "myositis with nemaline rods,"

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— not relevant to the inherited NEB disease.)

6. Mechanism / Pathophysiology

Ordered causal chain:

1. A **large heterozygous (or mosaic) deletion of 52–97 NEB exons** *leads to* loss of a substantial internal segment of nebulin and reduced/dysfunctional nebulin protein in the sarcomere. *(demonstrated:*

   P *40517164*

)
2. Reduced functional nebulin *results in* **failure of thin-filament length specification/ stabilization** (loss of molecular-ruler and CapZ-anchoring function). *(demonstrated:*

   P *35175800*

   P *18272787*

)

3. This *leads to* **shortened, non-uniform actin thin filaments** (~1.3 µm → ~0.75 µm in nebulin-deficient muscle). *(demonstrated:*

   P *19346529*

)
4. Shorter/non-uniform filaments *result in* **reduced thin–thick filament overlap and fewer force-generating crossbridges**, plus **reduced myofilament calcium sensitivity and altered crossbridge cycling kinetics**. *(demonstrated:*

   P *21115852*

   P *26123491*

)
5. In parallel (branch), reduced nebulin *leads to* **destabilization of the myosin super-relaxed (SRX) state**, *resulting in* **increased resting ATP consumption and remodeling of the energy proteome** (glycogen accumulation, abnormal mitochondria). *(demonstrated:*

   P *36528760*

)
6. In another branch, muscle-type-dependent **upregulation of proteolysis (MuRF-1/MUSA1/Gadd45a)** *results in* atrophy of glycolytic-fiber-rich muscles. *(demonstrated in mouse:*

   P *26123491*

)
7. Steps 4–6 converge to *cause* a **~60% reduction in force-generating capacity** and the **clinical phenotype of mild, distal-predominant muscle weakness** with nemaline rods, type 1 fiber predominance, and atrophy. *(demonstrated:*

   P *19346529*

   P *40517164*

)

Whether step 1→2 proceeds via haploinsufficiency or a dominant-negative poison effect of a truncated/internally-deleted nebulin is **inferred, not demonstrated**, for the dominant distal form.

Molecular/cellular detail: sarcomere thin-filament assembly and length regulation (GO:0030239 myofibril assembly; GO:0030240 skeletal muscle thin filament assembly; GO:0003009 skeletal muscle contraction); actin binding (GO:0003779); crossbridge cycling and ATP hydrolysis (GO:0030049 muscle filament sliding). **Cell type:** skeletal muscle fiber (CL:0000188 cell of skeletal muscle; CL:0008002 skeletal muscle fiber), with type 1 (slow oxidative) predominance. **Subcellular:** sarcomere / myofibril / Z-disc (GO:0030017 sarcomere;

GO:0031674 I band; GO:0030018 Z disc; GO:0005865 striated muscle thin filament); mitochondria secondarily abnormal. **Metabolic:** elevated resting ATP turnover from SRX loss; glycogen accumulation (CHEBI:28087). No primary immune involvement.

7. Anatomical Structures Affected

- **Organ/system:** skeletal muscle / musculoskeletal system (UBERON:0001134 skeletal muscle tissue; UBERON:0002385 muscle tissue). **Body systems:** musculoskeletal (primary); respiratory (secondary — respiratory muscles); bulbar/oropharyngeal (secondary).
- **Primary muscles:** distal lower-limb muscles — **soleus, gastrocnemius** (greatest fatty infiltration on MRI), tibialis; gluteus maximus also involved PMID: 40583855. UBERON:0011907 (gastrocnemius), UBERON:0001388 (soleus).
- **Tissue/cell:** striated skeletal muscle fiber (CL:0000188); type 1 fiber predominance.
- **Subcellular:** sarcomere thin filament, Z-disc (GO:0030017; GO:0030018).
- **Lateralization:** bilateral, broadly symmetric.

8. Temporal Development

- **Onset:** childhood to adult for the dominant distal form; **insidious/chronic**. (Congenital/neonatal onset characterizes severe recessive NEM2, not the dominant distal form.)
- **Progression: slow**, relatively **stable** course; distal-predominant; disease duration is chronic/lifelong PMID: 33397769; PMID: 40517164. MRI fatty infiltration correlates with disease duration ($r \approx 0.57–0.60$) PMID: 40583855.
- **Patterns:** progressive but non-episodic; no remission; critical intervention windows not defined.

9. Inheritance and Population

- **Inheritance: autosomal dominant** (single large heterozygous NEB deletion); mosaicism documented.
- **Epidemiology:** Nemaline myopathy overall ~**1 in 50,000 live births** (up to ~1 in 20,000 in some populations); NEB is the most common gene (>50%) PMID: 31530540. The dominant distal subtype is rarer; exact prevalence unknown (12 families in the defining study).
- **Penetrance/expressivity:** appears penetrant within affected families; expressivity variable (mild distal weakness).
- **Founder effects:** the exon-55 deletion is an Ashkenazi Jewish founder allele — but that is a **recessive** NEM2 allele, not the dominant deletion.

- **Carrier frequency/consanguinity:** relevant to recessive NEM2, not the dominant form. **Sex ratio:** no sex predilection expected for autosomal dominant inheritance.

10. Diagnostics

- **Muscle biopsy:** nemaline rods (Z-disc-derived electron-dense bodies) on light and electron microscopy; type 1 fiber predominance; ± cores (rod-core) PMID: 25205138; PMID: 42609597.
- **EMG:** myogenic (myopathic) pattern.
- **Muscle MRI:** distal-predominant fatty infiltration (soleus > gastrocnemius > gluteus maximus) PMID: 40583855.
- **Genetic testing:** NEB is very large and GC/repeat-rich, so **structural-variant-aware** approaches are essential — custom CGH arrays, short-read genome sequencing, droplet digital PCR, exome with CNV calling, and Sanger confirmation were all used to detect the large deletions PMID: 40517164. Standard exome panels can **miss** large NEB deletions; WGS/CMA/ddPCR improve detection.
- **Laboratory:** creatine kinase often normal or mildly elevated (nonspecific).
- **Differential diagnosis:** recessive NEM2; other thin-filament myopathies (ACTA1, TPM2, TPM3, TNNT1, KBTBD13, KLHL40/41); acquired SLONM (adult, monoclonal gammopathy); inflammatory myositis with rods. Distinguishing features: inheritance pattern, variant architecture, age/distribution of weakness, presence of monoclonal protein/inflammation.

11. Outcome / Prognosis

- The dominant distal form has a **favorable prognosis**: mild weakness, slow progression, generally preserved ambulation and life expectancy PMID: 40517164. This contrasts with severe recessive NEM2, which can cause neonatal respiratory failure and early death.
- Across the broader NM spectrum, morbidity includes mechanical support needs (58%), abnormal pulmonary function (65%), and bulbar dysfunction, but the course is *"relatively stable"* PMID: 33397769.
- **Prognostic factors:** variant architecture is the dominant determinant — large heterozygous deletions predict the mild distal phenotype PMID: 40517164. MRI fatty infiltration tracks disease duration.

12. Treatment

No curative therapy exists; management is supportive PMID: 38866581.

Modality	Intervention	NCIT suggestion
Respiratory	Non-invasive ventilation, pulmonary monitoring	NCIT:C116838 (respiratory support)
Physical/orthotic	Physiotherapy, ankle-foot orthoses for foot drop	NCIT:C15719 (physical therapy)
Nutritional/bulbar	Feeding support, swallowing management	NCIT:C15285 (nutritional support)
Orthopedic	Scoliosis/contracture management	NCIT:C51696 (orthopedic procedure)
Investigational (Ca^{2+} sensitizer)	Fast skeletal troponin activator (CK-2066260) — restores Ca^{2+} sensitivity in patient fibers [PMID: 23572184]	—
Investigational (myosin modulator)	Mavacamten — reverses ATP overconsumption in vitro [PMID: 39216086]	—

Ineffective/unproven: nutritional supplements — controlled model studies of L-tyrosine, L-carnitine, taurine, and creatine showed no significant functional benefit PMID: 29848386; PMID: 30065346. MuRF1 knockout increased muscle mass but not force PMID: 35897687. **Pharmacogenomics, gene/cell/RNA therapy, immunotherapy:** none established; gene replacement is impractical given nebulin's enormous size.

13. Prevention

- **Primary prevention:** not applicable (monogenic). **Genetic counseling** is the principal preventive tool — autosomal dominant transmission implies 50% offspring risk; mosaicism complicates recurrence-risk estimates PMID: 40517164.
- **Reproductive options:** prenatal testing and preimplantation genetic diagnosis are feasible once the familial deletion is defined.
- **Secondary/tertiary prevention:** surveillance of respiratory/bulbar function; orthotic and rehabilitation measures to prevent contractures and falls.
- **Immunization/public health:** not applicable.

14. Other Species / Natural Disease

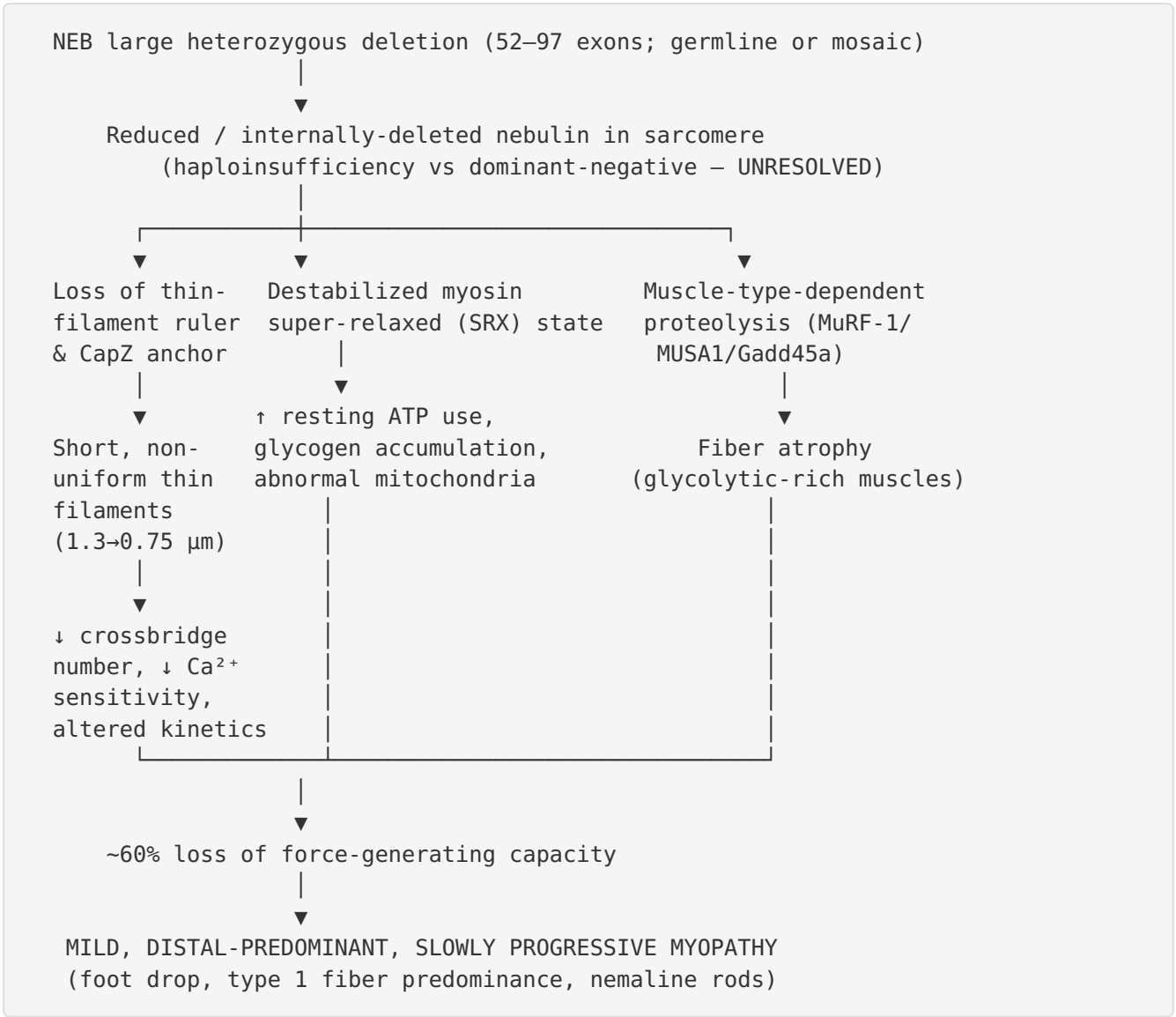
- **Model species with NEB orthologs:** *Mus musculus* (Neb), *Danio rerio* (neb), *Gallus gallus* (chicken nebulin, sequenced and conserved — 74–91% identity in the C-terminal region) PMID: 18522464.
- **Natural disease in companion animals/wildlife:** no well-documented naturally occurring NEB nemaline myopathy reported in OMIA for this specific entity.
- **Evolutionary conservation:** nebulin's modular repeat structure, Z-line binding, and actin-binding regions are conserved across vertebrates, supporting cross-species mechanistic validity.
- **Zoonotic potential:** not applicable (genetic disease).

15. Model Organisms

Model	Type	Key features / recapitulation	PMID
Conditional Neb-KO mouse (MCK-Cre)	Mammalian, conditional KO	<5% nebulin, nemaline rods, shortened thin filaments, reduced specific force, fiber-type switch, muscle-type-dependent atrophy	26123491
Neb ΔExon55 mouse	Knock-in (founder mutation)	Growth retardation, nemaline bodies, reduced thin-filament length and force	23715096
Novel Neb exon-55 deletion mouse	Knock-in	Recapitulates recurrent human variant	40108735
Zebrafish neb allelic series	Vertebrate	Reduced survival, defective muscle structure, reduced force, shorter thin filaments, Z-disc thickening	38493359
Patient-derived single myofibers	In vitro human	Force deficit, reduced Ca^{2+} sensitivity, SRX destabilization; used to test CK-2066260 and Mavacamten	23572184, 36528760

Limitations of models: most model the **recessive/loss-of-function** biology; a model specifically reproducing the **dominant large heterozygous deletion** (to test haploinsufficiency vs dominant-negative) is a key gap. **Resources:** MGI (mouse), ZFIN (zebrafish), Alliance of Genome Resources.

Mechanistic Model / Interpretation



The unifying interpretation is that **nebulin loss simultaneously corrupts thin-filament architecture and energetic homeostasis**. The thin-filament branch (shortening, reduced overlap, reduced Ca^{2+} sensitivity) is the primary driver of weakness; the SRX/energetic branch

explains the otherwise puzzling metabolic histology; and the proteolytic branch explains muscle-type-dependent atrophy. The dominant distal phenotype's **mildness** likely reflects the **heterozygous dosage** (residual nebulin from the intact allele) compared with the near-complete loss in biallelic recessive disease — a model consistent with haploinsufficiency, though the MONDO curation's assertion of a dominant-negative mechanism remains to be directly tested.

Evidence Base

PMID	Title (abbrev.)	Role
40517164	*Structural variation in nebulin... dominant distal phenotype*	**Defining study** — establishes dominant distal form from large heterozygous deletions
39802796	*(preprint of above)*	Preprint corroboration
35175800	*Structures from intact myofibrils... nebulin*	Molecular-ruler structural basis
19346529	*Thin filament length dysregulation...*	Quantifies thin-filament shortening (1.3→0.75 µm)
21115852	*Nebulin, a major player...*	Ca^{2+} sensitivity + crossbridge kinetics
26123491	*Nebulin deficiency in adult muscle...*	Conditional KO; force deficit + proteolysis
36528760	*NEB mutations disrupt super-relaxed state...*	Energetic/SRX branch
23572184	*Troponin activator augments force...*	CK-2066260 restores Ca^{2+} sensitivity
39216086	*Myosin ATPase inhibition fails to rescue...*	Mavacamten in vitro success / in vivo limit
23715096	*Deleting exon 55...*	ΔExon55 mouse; Ashkenazi founder
38493359	*Zebrafish allelic series...*	Vertebrate model recapitulation
25205138	*Mutation update: spectra of nebulin variants*	Phenotypic spectrum incl. distal
40583855	*Muscle MRI changes in NEB NM*	Distal MRI pattern
33397769	*Cross-sectional study of NM*	Natural history, stability
31530540	*Failure to identify modifiers...*	Epidemiology (1/50,000)
38866581	*L-tyrosine for infant with NM*	Supportive-only management
29848386	*Testing of therapies... lack of efficacy*	Supplements ineffective
42609597	*Cohort of 49 nebulin-related CM*	Histological spectrum confirmation
18272787	*Nebulin interacts with CapZ*	Z-disc anchoring

PMID	Title (abbrev.)	Role
26792324	*Nebulin, a multi-functional giant*	Multiple cellular roles

Evidence source types: human clinical/genetic cohorts (40517164, 40583855, 33397769, 42609597, 25205138); human in vitro single-fiber mechanics (19346529, 23572184, 36528760); model organism (26123491, 23715096, 40108735, 38493359, 39216086); structural/computational (35175800, 18272787).

Limitations and Knowledge Gaps

1. **Mechanism of dominance unresolved.** Whether large heterozygous NEB deletions cause disease by **haploinsufficiency** or a **dominant-negative** effect is not experimentally established for the dominant distal form. MONDO asserts dominant-negative, but no model specifically reproduces the heterozygous large deletion to test this.
2. **Small defining cohort.** The dominant distal entity is based on 12 families [PMID: 40517164]; precise prevalence, penetrance, expressivity, sex ratio, and long-term natural history are not yet quantified for this subtype specifically.
3. **Most functional/mechanistic data derive from recessive/KO models**, which may not fully capture the milder heterozygous biology.
4. **Detection challenges.** Large NEB deletions are easily missed by standard exome panels; the true frequency may be underestimated until SV-aware sequencing is routine.
5. **No disease-specific QoL or mortality datasets** for the dominant distal form.
6. **Therapeutics are preclinical.** Troponin activators and myosin modulators show in vitro promise but lack in vivo efficacy or clinical trial data for NEB disease.

Proposed Follow-up Experiments / Actions

1. **Build a heterozygous large-deletion model** (mouse or patient iPSC-derived myotubes) replicating a representative 52–97-exon deletion to directly test **haploinsufficiency vs dominant-negative** — the single highest-priority mechanistic question.

2. **Quantify nebulin protein and isoform composition** from the intact allele in patient muscle (Western/mass spec) to assess residual function and dosage effects.
3. **Establish a dominant-distal NEB natural-history registry** capturing onset, progression (serial MRI, MFM), respiratory/bulbar outcomes, penetrance, and QoL (PROMIS/SF-36).
4. **Advance fast skeletal troponin activators** toward trial-readiness in NEB-specific models, given their mechanistic fit (restore Ca^{2+} sensitivity in patient fibers).
5. **Test longer-duration / combination myosin modulation** (Mavacamten ± metabolic support) since 4-week in vivo treatment failed to correct the energy proteome [PMID: 39216086].
6. **Standardize SV-aware diagnostic pipelines** (WGS/ddPCR/CGH) for NEB to improve detection of large dominant deletions and refine epidemiology.
7. **Genetic counseling protocols** addressing mosaicism-adjusted recurrence risk for families with large NEB deletions.

Report compiled from 10 confirmed findings and 48 reviewed papers across 5 investigation iterations. Causal gene: NEB (nebulin), chromosome 2q23.3. Disease: MONDO:1010152, autosomal dominant nebulin-related myopathy.
